

Structural Validation Comparison Report

PCNA E113G Mutant Structure — PDB 3GPM

A comparative structural validation was performed before and after model refinement using **PROCHECK**, **ERRAT**, and **Verify3D**. The results show an overall improvement in several quality indicators, although Verify3D remains below its recommended threshold.

1. PROCHECK

The PROCHECK evaluation showed a clear reduction in structural errors after validation. **Errors decreased from 4 to 2**, while **pass evaluations increased from 2 to 3**. Warnings increased slightly from **2 to 3**, indicating that some residues still require attention, but the overall evaluation profile improved.

2. ERRAT

The **ERRAT Overall Quality Factor improved substantially from 72.8889 to 86.1905** following validation. This represents a notable improvement in the non-bonded interaction quality of the model and indicates a more favorable structural quality profile after refinement.

3. Verify3D

Verify3D showed a modest improvement in the percentage of residues achieving an average 3D–1D score ≥ 0.1 , increasing from **61.02% to 62.20%**. However, both models remained classified as **Fail**, since the recommended threshold of **80%** was not reached. Thus, despite the improvement, the model still has limitations in residue-environment compatibility.

Overall Assessment

The post-validation model demonstrates **measurable structural improvement**, particularly in ERRAT quality and PROCHECK error reduction. However, the persistent Verify3D failure indicates that further refinement may be beneficial. Overall, the validated structure has a **better quality profile than the pre-validation model**, but additional optimization should be considered before treating it as a fully validated structural model.